Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP Based Chatbot

R26-IT-018

Project Proposal Report

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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Traditional e-learning platforms rely heavily on textual and auditory instruction, creating significant cognitive barriers for deaf students who primarily utilize visual-spatial processing. Furthermore, these static systems lack the "emotional intelligence" to detect and mitigate a learner's frustration during complex subjects like G.C.E. O/L Information Technology. While existing adaptive learning systems adjust content based on post-assessment failure, and affective computing models focus heavily on neurotypical adults, a critical research gap exists in providing real-time, emotion-responsive interventions explicitly tailored to the visual needs of deaf teenagers.

This research proposes an emotion-aware adaptive e-learning system that actively monitors a deaf student's facial micro-expressions and interaction telemetry to dynamically modify educational content. Utilizing OpenCV for facial feature extraction and a TensorFlow-based Convolutional Neural Network (CNN) for emotion classification, the system calculates a real-time engagement score. When negative emotional states such as sustained confusion are detected, the Python backend triggers the React.js frontend to instantly transition dense textual UI into simplified, highly visual, analogy-based learning tiers retrieved from a MongoDB database.

The expected outcome is a seamless adaptation mechanism operating with under two seconds of latency, designed to prevent cognitive overload, reduce academic frustration, and significantly improve knowledge retention. Socially, this solution champions inclusive education by granting marginalized students equitable access to technical disciplines. Commercially, it establishes a scalable Software-as-a-Service framework viable for specialized educational institutions and inclusive mainstream schools across Sri Lanka.

Keywords: Affective Computing, Adaptive Learning, Deaf Education, Emotion Recognition, Inclusive E-Learning

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor, [Insert Supervisor's Name], for their invaluable guidance, continuous feedback, and expert mentorship throughout the development of this proposal. I also extend my appreciation to the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the academic foundation and resources necessary to pursue this research.

Finally, a deeply personal thank you goes to my father, my girlfriend, and my beloved daughter. Your unwavering support, endless patience, and constant encouragement during the long hours dedicated to this project have been my greatest motivation.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACE	Adaptive Content Engine
API	Application Programming Interface
CNN	Convolutional Neural Network
DHH	Deaf and Hard-of-Hearing
ERM	Emotion Recognition Module
FACS	Facial Action Coding System
HCI	Human-Computer Interaction
ITM	Interaction Tracking Module
LMS	Learning Management System
SaaS	Software as a Service
SPA	Single Page Application
UDL	Universal Design for Learning
WBS	Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

The foundational architectures of most modern educational technology platforms and Learning Management Systems are primarily designed with neurotypical, hearing students in mind. As global educational paradigms shift increasingly towards digital and remote learning environments, marginalized communities, particularly the deaf and hard-of-hearing, find themselves navigating systems that inherently disadvantage their cognitive learning styles. In the context of Sri Lanka, students preparing for their General Certificate of Education Ordinary Level examinations face immense academic pressure. When these students tackle highly technical and abstract subjects such as Information Technology, the reliance of traditional e-learning platforms on auditory cues, uncaptioned video lectures, and overwhelmingly dense textual explanations creates a profound barrier to comprehension.

Deaf learners do not simply learn exactly like hearing students minus the audio; they fundamentally rely on visual-spatial processing channels. Consequently, they benefit from instructional designs that prioritize high-quality visual aids, clear typography, and structured diagrammatic explanations over lengthy paragraphs. Furthermore, the most critical element missing from current e-learning platforms is the emotional intelligence inherently possessed by a physical human educator. In a traditional classroom setting, a special education teacher continuously monitors the non-verbal cues of their students. If a student exhibits a furrowed brow, a lack of focus, or signs of frustration, the teacher instinctively slows down the pace of the lesson, simplifies the vocabulary, or draws a diagram on the board to explain the concept through an alternative modality. Current digital learning environments are entirely blind to these emotional and cognitive struggles, leading to a static pedagogical experience that pushes deaf learners toward cognitive overload and eventual academic disengagement.

This research directly aligns with the United Nations Sustainable Development Goals, specifically Goal 4, which advocates for inclusive and equitable quality education, and Goal 10, which aims to reduce inequalities. By leveraging advancements in artificial intelligence and computer vision, specifically the sub-field of Affective Computing, it is now theoretically possible to engineer an educational platform capable of reading a student's facial expressions and interaction patterns. This creates a real-time feedback loop that dynamically adjusts the complexity and format of the learning material to suit the immediate cognitive and emotional needs of the marginalized learner.

1.2 Literature Review

Historically, the intersection of digital education and accessibility has been fraught with half measures, often limiting accommodations to simple video captioning rather than fundamental pedagogical restructuring. Research in accessible instructional design consistently emphasizes that deaf students achieve significantly higher comprehension and retention rates when educational content is supplemented with cognitive mapping tools and sequential visual guides. Scholars in the field of deaf education have documented that the working memory of deaf individuals is highly optimized for visual and spatial information, meaning that presenting complex Information Technology concepts through heavy blocks of text essentially forces these students to translate the information twice, drastically increasing their cognitive load.

Concurrently, the field of Affective Computing has gained significant traction within educational technology as researchers attempt to replicate the empathetic awareness of a human tutor. Utilizing deep learning frameworks, specifically Convolutional Neural Networks, modern systems have become highly adept at analyzing micro-expressions mapped to the universally recognized Facial Action Coding System. These intelligent tutoring systems can successfully determine states such as boredom, confusion, or high engagement. However, a critical review of the current literature reveals a substantial gap in the application of this technology. The vast majority of affective computing research in education has been calibrated for and tested on neurotypical university students. The application of real-time emotion recognition specifically tailored for deaf teenagers remains largely unexplored.

Furthermore, adaptive learning systems currently available in the commercial and academic spheres largely rely on a concept known as knowledge tracing. In these systems, the platform adapts the curriculum based strictly on the student's performance metrics, such as quiz scores or the number of incorrect attempts on an assignment. While this is an effective method for assessing academic competence, it completely fails to capture the student's emotional journey. A student might eventually answer a programming question correctly but experience severe frustration and anxiety in the process. Emotion-aware adaptation introduces a necessary paradigm shift. By modifying the content before the student fails an assessment, the system maintains the learner in a state of productive engagement, preventing the emotional burnout that frequently leads to high dropout rates among students with disabilities.

1.3 Research Gap

While technology has advanced, there is still a significant disconnect in how we teach technical subjects like Information Technology to deaf students in Sri Lanka. Current e-learning tools often fail these learners because they are designed for hearing students, relying on heavy text that creates a massive "cognitive load" for those who process information visually. Our research addresses several critical gaps

- **Reactive vs. Proactive Support**: Most existing systems only notice a student is struggling after they fail a quiz. There is a lack of platforms that intervene *the moment* a student looks confused.
- **The "Silent" Barrier**: Traditional platforms like Moodle often provide only static captions, which doesn't help a student who is struggling with abstract IT concepts like networking or logic gates.
- **Lack of Emotional Intelligence**: A human teacher can see a student's "furrowed brow" and instinctively slow down. Digital tools today are effectively "blind" to these vital non-verbal cues.
- **Localized Curriculum**: Most affective computing research is tested on university students abroad, leaving a gap for specialized support for Sri Lankan G.C.E. O/L teenagers

Comparison of Research with the Proposed Solution

Feature	Traditional LMS	General AI Research	Proposed System
Integrated Emotion + Visuals	No	Partial	Yes
G.C.E. O/L IT Syllabus Focus	No	No	Yes
Under 2-Second Adaptation	No	No	Yes
Tiered Visual Materials	No	No	Yes

1.4 Novel Contribution

The proposed research introduces a **multi-modal, emotion-aware adaptation framework** designed specifically to bridge the gap between affective computing and deaf education. Key contributions include:

- **Real-Time Emotional Intervention**: The system introduces a seamless integration of **facial micro-expression analysis** and interaction telemetry to trigger pedagogical shifts within **under two seconds** of latency.
- **Hierarchical Content Tiering**: A novel approach to content delivery that maps the IT syllabus into **three distinct tiers** moving from standard text to highly visual, analogy-based interfaces automatically based on student frustration levels.
- **Privacy-First Biometric Processing**: Unlike many systems that require cloud-based video processing, this research contributes a framework where **facial landmark extraction** occurs entirely on the client side, ensuring student privacy while maintaining high-fidelity emotion recognition.
- **Inclusive SaaS Framework**: It establishes a scalable **Software-as-a-Service (SaaS)** model specifically optimized for the Sri Lankan inclusive education landscape.

1.5 Research Problem

The core focus of this research is the development of an **Emotion-Aware Adaptive Content Engine (ACE)** that can simulate the empathetic awareness of a human special education teacher. Deaf students often face "cognitive barriers" when navigating static, text-dense platforms. Therefore, the research problem focuses on several technical and pedagogical challenges:

1. **Latency and Immersion**: How to maintain a real-time feedback loop that classifies emotions and re-renders the **Document Object Model (DOM)** without disruptive page reloads that break the student's focus.
2. **Semantic Mapping of Technical Content**: The challenge of automatically simplifying complex IT concepts (like logic gates or networking) into **Tier 3 (Highly Visual Analogies)** while maintaining academic accuracy for the O/L examination.
3. **Accuracy in Diverse Environments**: Engineering a **Convolutional Neural Network (CNN)** that can achieve over **85% accuracy** in classifying concentration, confusion, and frustration among teenagers in varied lighting and hardware conditions.
4. **Graceful Degradation**: Ensuring the system remains functional and supportive via **Interaction Telemetry** (mouse movement/scrolling) even if the user denies webcam access or experiences hardware failure.

2. OBJECTIVE

2.1 Main Objective

The primary objective of this research is to design, develop, and scientifically validate an emotion-aware, adaptive e-learning module tailored specifically for deaf students studying Information Technology at the G.C.E. Ordinary Level. This system will actively monitor Real time emotional and engagement metrics to dynamically modify the complexity, format, and visual density of educational content.

2.2 Specific Objectives

1. **Designing Hierarchical Content Tiers**: Collaborate with special education experts to create three distinct levels of learning materials mapped to the national G.C.E. O/L Information Technology syllabus:

- **Tier 1**: Standard textual explanations.
- **Tier 2**: Simplified content with reduced linguistic complexity.
- **Tier 3**: Highly specialized, visually dominant interfaces featuring interactive analogies.

2. **Developing a Rule-Based Adaptive Algorithm**: Build a sophisticated engine that synthesizes emotional data and interaction metrics (like mouse movements and scrolling velocity) to automatically trigger transitions between content tiers without requiring manual student input.
3. **Evaluating Pedagogical Effectiveness**: Conduct rigorous user testing with deaf teenage learners to evaluate the system's technical usability and impact on engagement, comparing their knowledge retention and comprehension rates against a control group using a static learning environment.

3. METHODOLOGY

3.1 Research Approach

This project follows the **Design Science Research (DSR)** methodology, which is the gold standard for creating innovative IT solutions that solve real-world social challenges. Rather than just studying a problem, DSR focuses on building and testing a "solution artifact"—in this case, an **Emotion-Aware Adaptive E-Learning Platform** designed to help deaf students master technical IT concepts.

The research process is organized into a rigorous cycle of design and evaluation:

- **Identifying the Problem**: We start by looking at the "cognitive barriers" deaf students face when trying to learn abstract G.C.E. O/L IT topics using platforms that rely too heavily on text.
- **Defining Requirements**: We analyze the need for a system that can "see" a student's frustration and respond instantly, much like a human teacher would in a classroom.
- **Designing the Artifact**: We've designed a multi-layered architecture where the **Emotion Recognition Module (ERM)** and **Interaction Tracking Module (ITM)** provide the data needed to make smart teaching decisions.
- **Development and Implementation**: The system is built using a modern technical stack—**React.js** for a smooth user interface and **Python** for the heavy-lifting AI tasks like emotion classification.
- **Rigorous Evaluation**: To prove the system actually works, we will conduct user testing with deaf teenage learners. We'll compare their progress and engagement levels against a group using a traditional, static platform to measure the real-world impact.

The Responsive Feedback Loop

The core of this research is a continuous, closed-loop feedback mechanism. The **Adaptive Content Engine** acts as the "brain" of the platform, taking in real-time emotional cues and interaction data to humanize the digital experience. If the system detects a student is stuck, it doesn't wait for them to fail; it proactively swaps out confusing text for **Tier 3 visual analogies** to keep them engaged and moving forward.

3.2 High-Level System Architecture and Data Flow

The proposed emotion-aware e-learning system operates on a continuous, closed-loop feedback architecture designed to monitor, analyze, and adapt the interface in real-time. The architecture is logically divided into three primary interconnected subsystems known as the Emotion Recognition Module, the Interaction Tracking Module, and the Adaptive Content Engine.

During the initial phase of user interaction, the presentation layer captures two distinct streams of data through the web browser. The first stream is a live video feed obtained via the user's webcam, requiring explicit, informed consent prior to activation. The second stream consists of interaction telemetry, which quietly monitors the user's mouse movements, scrolling velocity, click frequency, and periods of absolute idle time.

These data streams are securely routed to the processing layer. The video feed undergoes immediate analysis where computer vision techniques extract precise facial landmarks, tracking the geometric relationships between the user's eyes, eyebrows, and mouth. Simultaneously, the interaction telemetry is mathematically weighted to generate a continuous numerical value representing the user's physical engagement with the page.

The extracted emotional classification and the calculated engagement score are then passed directly into the decision layer, managed by the Adaptive Content Engine. This engine utilizes a defined set of logical rules to determine the current cognitive state of the student. If the algorithm determines that a pedagogical intervention is required due to sustained negative emotions, it sends a payload back to the frontend presentation layer. The user interface then dynamically re-renders the Document Object Model, instantly replacing the confusing dense textual content with simplified, highly visual alternative materials.

3.3 Technical Implementation Details

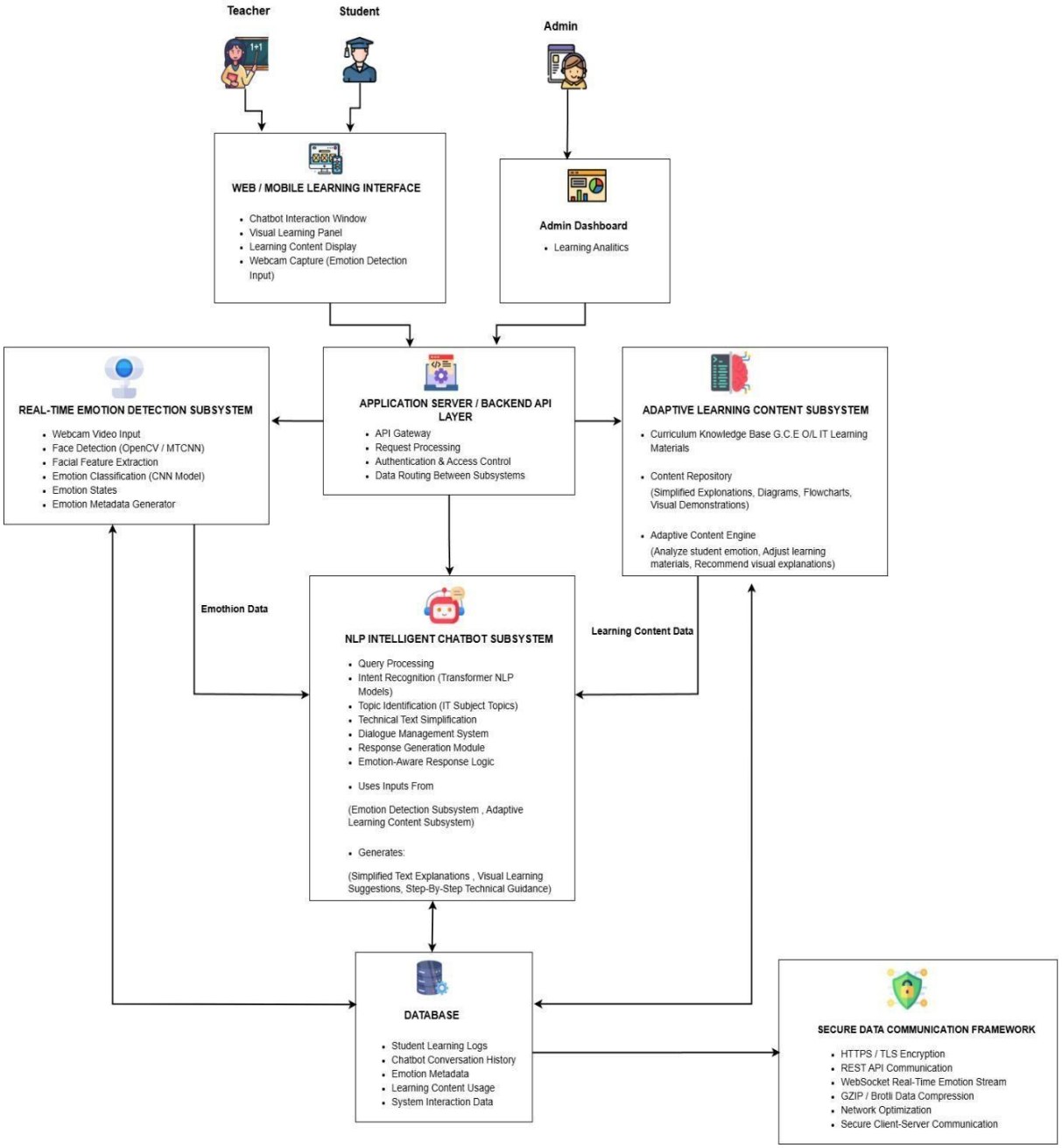
To ensure real-time responsiveness and seamless content adaptation without disruptive page reloads, a modern web development stack will be strictly utilized. The frontend environment will be constructed using React, as its inherent state management capabilities are perfectly suited for instantly swapping interface components based on backend triggers.

The backend processing environment will be built using Python, selected specifically for its native, highly optimized compatibility with industry-standard machine learning libraries. The facial feature extraction and frame processing will be handled by OpenCV, while the deep learning model responsible for emotion classification will be constructed using TensorFlow and Keras. This Convolutional Neural Network will be fine-tuned using standardized, opensource facial expression datasets to ensure baseline accuracy before being tested on the target demographic.

All unstructured interaction logs, user session data, and the massive array of hierarchical educational assets will be stored in a NoSQL database system, specifically MongoDB, allowing the adaptive engine to rapidly query and retrieve visual assets based on the student's shifting emotional state.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Figure 1: high Level System Architecture



4.1 System Components and Interaction

The high-level architecture of the proposed emotion-aware e-learning system is built upon a distributed, highly cohesive framework that separates the user interface from the heavy computational demands of machine learning inference. At the forefront is the presentation layer, constructed as a dynamic single-page application. This layer is responsible for rendering the educational content, managing state transitions between different pedagogical tiers, and securely capturing the user's webcam feed and interaction telemetry. The webcam feed is processed locally within the browser to the greatest extent possible using lightweight computer vision libraries to extract facial landmark coordinates, ensuring that raw video data is never transmitted across the network, thereby preserving user privacy.

These extracted coordinates, alongside the calculated engagement metrics such as cursor velocity and idle duration, form a continuous stream of structured data that is transmitted via secure WebSockets to the backend processing layer. The backend architecture is built utilizing a high-performance Python framework, acting as the central nervous system of the application. Upon receiving the incoming data stream, the backend routes the facial coordinates through a pre-trained Convolutional Neural Network. This deep learning model is specifically optimized to classify the subtle micro-expressions of the student into predefined cognitive states, primarily focusing on concentration, confusion, and frustration.

Simultaneously, the incoming interaction telemetry is evaluated by an algorithmic scoring system to generate an overarching engagement score. The classified emotional state and the numerical engagement score are then synthesized by the core component of the backend, the Adaptive Content Engine. This engine evaluates the combined data against a rigorous set of pedagogical rules. When the engine detects a sustained period of negative emotion or a drastic drop in the engagement score, it generates an adaptation payload. This payload contains instructions detailing exactly which tier of visual content should be displayed. The payload is transmitted back to the presentation layer, which seamlessly updates the Document Object Model to present the simplified, highly visual learning materials without interrupting the student's workflow with disruptive page reloads.

4.2 Scalability and Security Considerations

In terms of scalability, the system is designed to leverage cloud-based infrastructure to handle concurrent user sessions dynamically. The backend services, particularly the machine learning inference engine and the Adaptive Content Engine, are containerized. This microservices approach allows the platform to automatically provision additional computational resources during peak usage hours, ensuring that the latency between emotion detection and content adaptation remains strictly under the two-second threshold necessary for maintaining immersion. The database architecture utilizes a NoSQL document-oriented structure, which provides the flexibility required to store highly varied and unstructured educational assets, ranging from standard text paragraphs to complex interactive visual

analogies, while also supporting rapid read operations critical for real-time interface adaptation.

Security and ethical compliance form the foundational pillars of this architecture, particularly given the involvement of marginalized teenage users and the utilization of biometric data. The system enforces strict data minimization protocols. Raw video feeds are explicitly prohibited from being recorded, stored, or transmitted to any external server. The initial facial feature extraction occurs entirely on the client side, and only the mathematically abstracted coordinate data is sent to the backend for emotion classification. Furthermore, all data transmitted between the client interface and the backend servers is encrypted using industry standard transport layer security protocols. User profiles and session logs stored in the database are anonymized and decoupled from personally identifiable information, ensuring that a student's emotional journey and academic struggles remain entirely confidential and protected against unauthorized access.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The successful development and deployment of this adaptive e-learning platform depend on fulfilling the complex needs of several primary and secondary stakeholders. The primary stakeholders are the deaf and hard-of-hearing students undertaking the Sri Lankan G.C.E. Ordinary Level Information Technology curriculum. These individuals require a digital learning environment that respects their visual-spatial cognitive processing styles and actively mitigates the frustration induced by traditional, text-dense platforms. The secondary stakeholders comprise special education teachers and instructional designers. These professionals require a robust technological framework that allows them to map existing national syllabi into hierarchical, visually driven content tiers, effectively digitizing the empathetic interventions they would normally perform in a physical classroom. The tertiary stakeholders include the Ministry of Education and regional curriculum developers, who are invested in scalable, accessible technologies that can improve national examination pass rates among marginalized student demographics and ensure compliance with inclusive education policies.

5.2 Functional Requirements

Functional requirements define the specific behaviors, tasks, and actions the system must be able to perform to operate successfully.

1. **Webcam Video Capture:** The system must request explicit user permission to access the local webcam and capture a continuous live video feed during active learning sessions.
2. **Real-Time Emotion Classification:** The Emotion Recognition Module (ERM) must process the incoming video frames to extract facial landmarks and classify the user's emotional state into one of four primary categories: Focused, Confused, Frustrated, or Bored.
3. **Interaction Telemetry Tracking:** The Interaction Tracking Module (ITM) must actively monitor and record the user's mouse movements, scrolling velocity, click frequency, and periods of absolute idle time.
4. **Engagement Score Calculation:** The system must utilize the data gathered by the ITM to continuously calculate and update a numerical "Engagement Score" ranging from 0 to 100.
5. **Hierarchical Content Management:** The database must store G.C.E. O/L Information Technology syllabus content mapped into three distinct tiers: Tier 1 (Standard Text), Tier 2 (Simplified Bullet Points), and Tier 3 (Highly Visual Analogies).
6. **Automated Content Adaptation:** The Adaptive Content Engine (ACE) must

automatically trigger a UI update, transitioning the content to a lower visual tier, if the system detects sustained "Confusion" or "Frustration" for a period exceeding 15 seconds.

7. **Dynamic DOM Re-rendering:** The frontend application must seamlessly swap out text-heavy components for visual diagrams without requiring a full page reload, preserving the user's current scroll position.
8. **Manual User Override:** The system must provide a persistent, accessible control mechanism allowing the student to manually override the automated adaptation and return to standard textual content at any time.
9. **Educator Analytics Dashboard:** The platform must provide a secure portal for special education teachers to view anonymized, aggregated reports on class-wide emotional engagement and identify syllabus topics that consistently trigger high frustration levels.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance goals, and security standards the system must meet to ensure a viable user experience.

- **Performance and Latency:** The end-to-end process—from the initial physical expression of frustration to the final rendering of the adapted visual content on the screen—must execute with a maximum latency of strictly under two seconds to maintain user immersion.
- **Algorithmic Accuracy:** The Convolutional Neural Network utilized for facial emotion classification must achieve a minimum baseline accuracy of 85% when evaluated against standardized facial expression testing datasets.
- **Accessibility and Usability:** The user interface must strictly comply with Web Content Accessibility Guidelines (WCAG) tailored for deaf users. This includes utilizing high-contrast color schemes, maintaining a minimum sans-serif font size of 16px, and completely eliminating any reliance on audio-based alerts or prompts.
- **Data Privacy and Biometric Security:** The system must enforce transient data processing protocols. Raw video footage must never be recorded, transmitted across external networks, or stored in the database. Only mathematically abstracted facial coordinate data may be sent to the backend for inference.
- **Data Encryption:** All telemetry data, engagement scores, and emotional classifications transmitted between the React frontend and the Python backend must be encrypted using standard Transport Layer Security (TLS).
- **System Reliability and Fallback:** In the event that a user denies webcam permissions, or if local hardware fails, the system must gracefully degrade and rely entirely on the Interaction Tracking Module (mouse/scroll data) to drive the Adaptive Content Engine, ensuring the student can still access the educational materials.
- **Scalability:** The backend inference engine must be containerized and capable of horizontally scaling through cloud instances to handle concurrent traffic spikes during standard school hours without degrading the two-second latency threshold.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed Emotion-Aware E-Learning System includes educational institutions and organizations that support deaf and hard-of-hearing students. This includes special education schools, inclusive schools, and educational training centers that provide learning support for students with hearing impairments.

The system is specifically designed for students studying the G.C.E. O/L IT syllabus**,** where technical concepts often require clear explanations and visual learning materials.

A typical customer persona is a 15–16-year-old deaf student preparing for the G.C.E. O/L IT examination**.** These students prefer visual learning methods such as diagrams, simplified explanations, and step-by-step demonstrations instead of long text-based lessons.

Secondary users include teachers and school administrators who want to improve learning accessibility and monitor student learning progress through analytics.

6.2 Value Proposition

The proposed system provides an intelligent learning platform specifically designed to improve accessibility for deaf students. Traditional e-learning systems rely heavily on text or audio explanations, which can be difficult for deaf learners to understand.

This platform offers several key benefits:

- **Emotion-aware learning support** that detects student emotions during learning sessions.
- **Adaptive learning content** that adjusts explanations and visual materials based on the student's understanding.
- **Interactive AI chatbot tutor** that answers student questions using simplified explanations.
- **Visual learning materials** such as diagrams, flowcharts, and step-by-step demonstrations.

By combining emotion detection, adaptive learning, and NLP-based chatbot support, the system creates a more personalized and inclusive learning environment for deaf students.

6.3 Revenue Model

The system can generate revenue through several simple models suitable for an educational platform.

- **Subscription Model:**
  Educational institutions can subscribe to the platform by paying a yearly or monthly fee to allow their students to use the system.
- **Institutional Licensing:**
  Schools or educational centers can purchase a license that allows multiple students and teachers to access the system.
- **Freemium Model:**
  Basic learning support and chatbot features can be provided for free, while advanced features such as detailed learning analytics, additional visual resources, and advanced AI support can be included in a premium plan.

6.4 Cost Estimation

The development and operational costs of the system include several components as shown below.

Table 1: Cost Estimation

Category	**Estimated Cost (Initial Year)**	**Description**
Development Tools	Low / Free	Use of open-source frameworks such as Python, OpenCV, TensorFlow, and NLP libraries
Cloud Hosting	$60 – $80	Basic cloud server for backend API and database
Learning Content Production	$20 - $50	Creating diagrams and visual learning materials
Database Services	$30 – $50	Hosting MongoDB database
System Maintenance	$50 - $80	System updates and bug fixes

6.5 Pricing Strategy

The pricing strategy focuses on affordability so that educational institutions and students can easily adopt the system.

Possible pricing options include:

1. **Free Basic Plan:** Limited chatbot interaction and basic learning support.
2. **Student Plan:** Approximately $3–$5 per month for full access to chatbot tutoring and adaptive learning materials.
3. **Institutional Plan:** Approximately $200–$500 per year for schools that want to provide access to multiple students.

This pricing model ensures that the platform remains accessible while still providing sustainable funding for system maintenance and improvements.

6.6 Competitive Advantage

The proposed system offers several advantages compared with traditional e-learning platforms.

First, the system is specifically designed for deaf and hard-of-hearing students**,** focusing on visual explanations and simplified learning content.

Second, the integration of emotion detection technology allows the system to understand student emotional states and adjust learning materials accordingly.

Third, the system combines emotion detection, adaptive learning, and chatbot tutoring in a single platform, which is not commonly available in traditional e-learning systems.

These features provide a more personalized and engaging learning experience for students.

6.7 Intellectual Property Considerations

The proposed system includes several components that can be protected as intellectual property.

The software architecture, chatbot algorithms, and system implementation can be protected through software copyright protection**.** Additionally, custom-developed learning materials, diagrams, and educational content can also be protected under copyright laws.

If the system demonstrates significant novelty, future research may explore applying for patent protection for the emotion-aware adaptive learning methodology used in the platform.

Protecting the intellectual property of the system will help ensure ownership of the developed technology and support potential commercialization opportunities in the future.

7. BUDGET AND JUSTIFICATION

The development of the proposed Emotion-Aware E-Learning System requires several resources including hardware devices, software tools, cloud infrastructure, and human development effort. Although the project is developed within a university environment, it is important to estimate the actual cost of system development to understand the resources required if the system is implemented in a real-world environment. The following sections present the estimated budget and justification for each cost category.

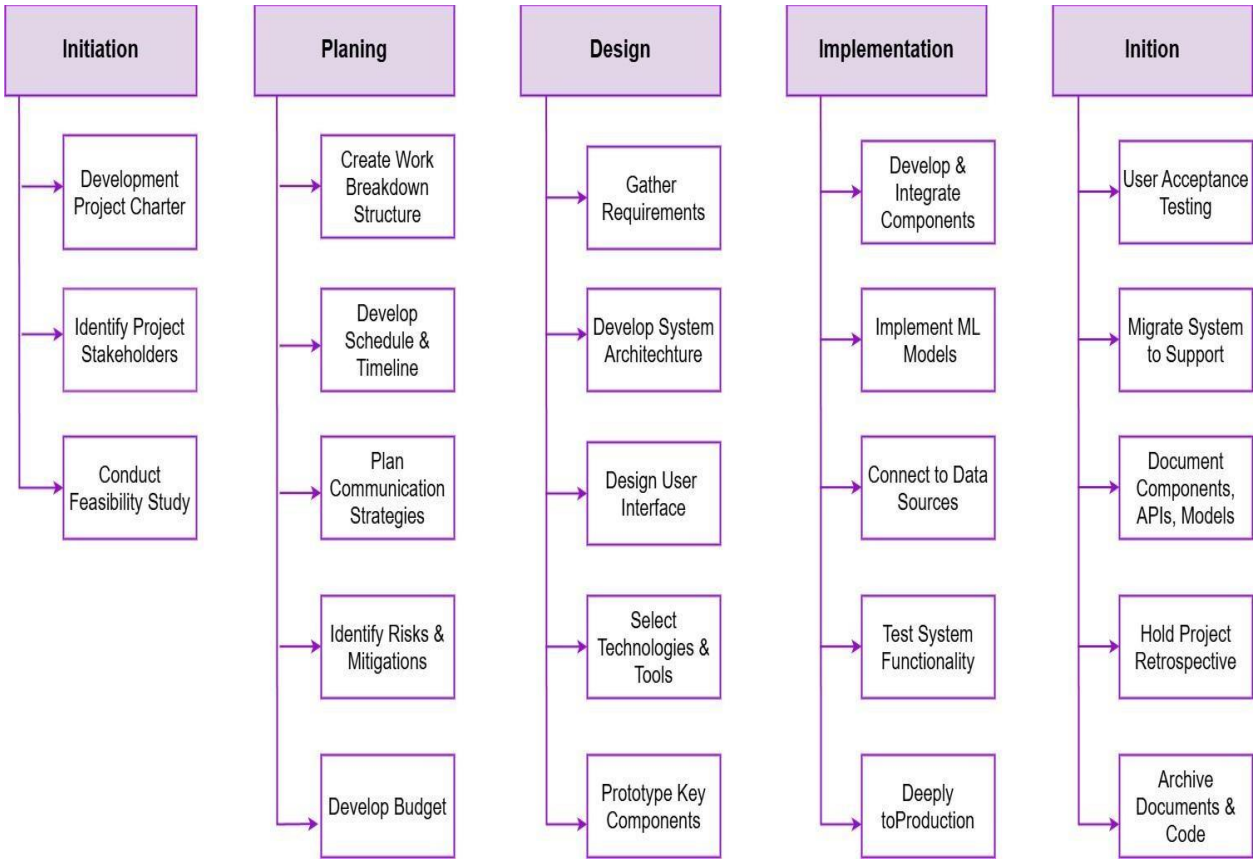
Table 2: budget

Category	**Item**	**Estimated Cost (LKR)**
Hardware	Laptop / Development Computer	250,000
Hardware	Webcam	8,000
Hardware	Internet Connectivity	6,000
Software	Open-source Development Tools (Python, OpenCV, TensorFlow, NLP Libraries)	0
Cloud Services	Cloud Hosting and Database Services	15,000
Human Effort	AI Model Development and Chatbot Implementation	200,000
Human Effort	Web Application Development and System Integration	120,000
Human Effort	System Testing and Evaluation	60,000
Total Estimated Budget		**659,000**

The budget mainly includes hardware resources, cloud services, and human development effort required to implement the system. A laptop is required for AI model development, software programming, and system testing. A webcam is necessary for capturing facial expressions used in the emotion detection subsystem. Cloud services are required to host the backend server, database, and learning materials so that the system can be accessed online. Human effort represents the estimated cost of software development, including AI model implementation, chatbot development, web application development, and system testing. Open-source technologies such as Python, TensorFlow, OpenCV, and NLP libraries are used to reduce software licensing costs while still enabling advanced AI-based functionality.

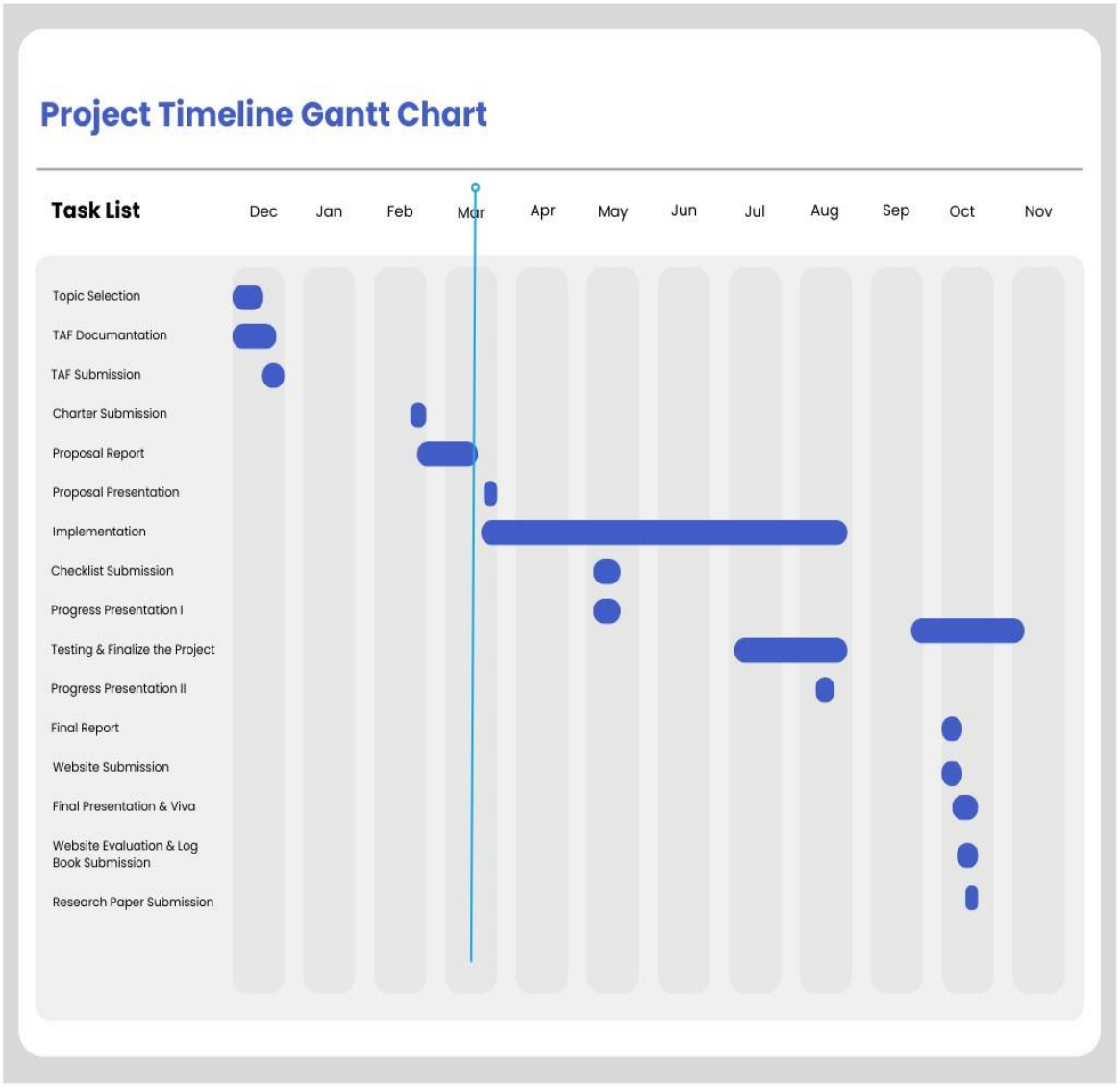
8. WORK BREAKDOWN STRUCTURE (WBS)

Figure 2: Work Breakdown Architecture



9. GANTT CHART

Figure 3: Grantt Chart



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